The Impact of Spatiality and Modality on Group Communications in Extended Reality Teleconferencing

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To the Directors of the Program on Symbolic Systems:

I certify that I have read the thesis of Violet Crow

in its final form for submission and have found it to be satisfactory for the degree of

Bachelor of Science with Honors.

[Electronically signed]

June 5th, 2026

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Abstract

Conversation is impacted by the spatial relationship of interlocutors and how they perceive their surroundings. Yet, due to challenges in isolating these variables, there has been little academic research on the topic. New technologies now allow researchers to test high-fidelity, photorealistic avatars that track and represent real-time facial expressions and head and hand movements for users wearing headsets. The current study examines groups of two, three, or four individuals as they collaborate on a verbal task wearing such headsets. 133 participants were assigned to one of four conditions in a 2 (spatiality: spatial versus grid) by 2 (modality: virtual reality versus mixed reality) factorial design to examine verbal, nonverbal, and self-report measures. Results show that participants attend more to the faces of others in the grid configuration than in the spatial configuration, and that those in fully immersive virtual reality consistently paid less attention to the faces of their group members than those in mixed reality passthrough. Contrary to expectations, the spatial representations and environment modality did not mitigate subjective experiences of cognitive or physical task load. Instead, group size emerged as the primary factor shaping subjective social outcomes, with larger groups reporting significantly lower entitativity and higher feelings of ostracism. We discuss the practical implications for designing collaborative systems, as well as theoretical contributions to the literature on conversational dynamics and interpersonal attention.

The Impact of Spatiality and Modality on Group Communications in Extended Reality Teleconferencing

Imagine joining a weekly remote meeting of global team members. Rather than seeing a flat, two-dimensional grid of webcams on a laptop screen, however, you are joining using a mixed reality (MR) headset. Photorealistic, three-dimensional avatars of your coworkers appear to sit across your physical coffee table. Their speech originates from their location in the room, and when they gesture to someone, it is coherent to all. You can even share mutual eye contact when someone makes a bad joke. Despite joining from across the globe, each person in the meeting feels as if the whole team is together in one physical space. While this spatial interface creates a sense of "being there," it also introduces new complexities. How are the norms of conversation and collaboration impacted by avatars that look convincingly human and seem to occupy your physical space, but are not actually there?

Much has been written about the causes of “Zoom fatigue” (Bailenson, 2021) and how the mechanics of standard teleconferencing may contribute to negative psychological outcomes (Johnson, 2021), hinder creative idea generation in groups (Brucks & Levav, 2022), and disrupt group communication patterns (Kim et al., 2025). Driven by the mass adoption of remote work, teleconferencing has become defined by the default “gallery view” configuration (Balogová & Brumby, 2022). However, a boom in demand-driven teleconferencing innovations (Davis et al., 2026) has resulted in new configurations that more closely resemble the spatial dynamics of in-person conversations, successfully decreasing stress and cognitive load (Baym et al., 2021). While these spatial-resemblant configurations are far preferred by users to the gallery view (Tang et al., 2023), their benefits are constrained by the use of two-dimensional screens and webcams. Today, virtual reality (VR) and MR technologies utilize advanced placement retargeting algorithms to preserve the spatial semantics of teleconferencing avatars across dissimilar physical environments, effectively blending physical and virtual spaces (Okhovvat et al., 2024). Yet, the psychological trade-offs of imposing physical-world social norms onto mediated avatars remain under-researched.

Previous research indicates that both traditional two-dimensional and spatial three-dimensional configurations can influence social dynamics in groups. Standard remote communication systems can disrupt natural communication patterns and exacerbate differences in member influence in groups (Kim et al., 2025), potentially leading to poor entitativity (the perception of a group as a cohesive entity) and feelings of isolation. Conversely, embodying spatial avatars that preserve physical-world semantics may foster a stronger shared identity among group members as posited by Han and colleagues (2023), and having higher feelings of group belongingness has been tied to lower post-videoconference fatigue (Bennett et al., 2021).

Extended reality (XR) teleconferencing offers a sense of co-presence, or sharing a physical space with another individual, which has the potential to be stronger than standard teleconferencing (Yousefdeh & Oyelere, 2024). However, there remain gaps in the literature.

First and foremost, until very recently it has not been possible to use avatars that are photorealistic, highly resemblant to their users, and can track real-time facial expressions and head and hand movements. While networked avatar systems have been implemented previously in VR (see Han et al., 2022 for a review), there is little work implementing and studying networked avatars in MR. Technology capable of both appropriately situating teleconferencing avatars in mixed reality and rendering them in VR with comparable fidelity to MR, is relatively new. Consequently, there is little work comparing fully immersive VR with non-immersive MR (Heiskanen, 2022), particularly for teleconferencing applications, that does not also involve significant confounds.

To address the intersection of spatiality, avatar fidelity, and environment virtuality, the current study investigates how these factors influence group dynamics as measured by verbal and nonverbal behavior, social cohesion, and cognitive load. Using the Apple Vision Pro, groups of two to four participants engaged in a verbal collaboration using an adapted, unmoderated Desert Survival consensus-building exercise. The communication interface was manipulated in a 2x2 between-subjects factorial design, altering both the *Spatiality* of the interlocutors (avatars rendered in three-dimensional space vs. avatar windows grouped in a grid) and the background *Modality* (mixed reality passthrough vs. fully immersive virtual reality). By isolating these architectural affordances, we aim to answer questions in the following areas:

RQ1. How do spatiality (avatars situated spatially vs. in a grid), modality (MR vs. VR), and group size in teleconferencing affect group-level social dynamics such as co-presence, group cohesion, entitativity, and ostracization?

RQ2. How do spatiality, modality, and group size affect communication behaviors, including self-reported collaboration dynamics and the use of affiliation-related words and collectively-referential pronouns?

RQ3. How do spatiality, modality, and group size affect the distribution of visual attention toward interlocutors and the surrounding environment, as measured by analysis of point-of-view and face detection?

Traditional two-dimensional teleconferencing has been associated with cognitive exhaustion (Bailenson, 2021; Johnson, 2021), while spatial configurations have been shown to decrease stress and cognitive load (Baym et al., 2021). Building on this, we pose the following research question and hypothesis:

RQ4. How do spatiality and modality affect a participant's feelings of physical load, cognitive load, and emotional discomfort?

H1: Participants who see interlocutors spatially will report lower levels of physical load, cognitive load, and emotional discomfort than those in standard grid configurations, as predicted by survey data from previous work (Fauville et al., 2023).

Related Work

Groups in XR

The Impact of Group Size. Group size is known to be consequential for face-to-face meetings (Allen et al, 2021) and teleconferencing (Rogelberg, 2020). In non-VR contexts, larger group sizes are associated with reductions in performance due to difficulties in reaching a consensus and increased free riding (de Back et al., 2021). However, the number of studies investigating groups larger than dyads in XR is quite slim (Han et al., 2022), despite the vastly different dynamics possible in larger groups.

With more than one possible referent in a conversation, nonverbal behaviors like eye contact, head orientation, and other spatially referential gestures become essential to indicating attention, which can negotiate turn-taking (Bailenson et al., 2002; Roth et al., 2018; Smith & Neff, 2018). The impact of spatiality on turn-taking and attentional signaling could thereby be mediated by group size.

Much is known about the impact of avatar realism as well, relevant given the unprecedented photorealism provided by the avatars used in this study, but most of the knowledge is limited to dyads. We know that photorealistic, highly expressive avatars with floating hands are highly preferred and lead to better quality interactions in many facets than less realistic ones (Aseeri & Interrante, 2021; De Simone et al., 2019; Herrera et al., 2020; Yoon et al., 2020; Garau et al., 2003; Wu et al., 2021). By providing all participants with avatars of the highest visual and behavioral fidelity, including floating but realistic tracked hands and fully tracked facial features, we minimize the detrimental effects avatars may have compared to real face-to-face communication.

The impact of location is relevant as well; we know that physical separation increases presence and performance during collaborative VR (Born et al., 2019) and that perceptual conflicts around avatar location impact performance in VR while preventing overlap enhances it (Chen et al., 2014; Hoppe et al., 2020). In this study, participants were physically separated from each other to increase presence and performance, and the native features of the interface maintained mutually comprehensible spatial relationships between participants, removing the possibility of perceptual conflicts and overlap.

Additionally, it is known for dyads that conveying spatial awareness via avatars impacts mental load and head movements (Irlitti et al., 2019).

It is unknown which of these dyadic effects are consistent for larger groups, such as the groups of three and four in this study. For groups of three, VR was found to have higher general, emotional, social, and visual fatigue compared to two-dimensional videoconferencing and face to face interactions on a task similar to the one used in this study (Frontzkowski et. al, 2025). Groups of four have been found to better support interaction and collaboration in VR-based learning compared to larger groups (Chen et al., 2026). Both of these findings motivate our investigation of these factors.

The current study is the first to examine triads and tetrads in both VR and MR teleconferencing with no confounds of environmental fidelity, and with all participants using immersive headsets. It is also one of few studies to study groups larger than two in XR interactions exclusively without confederates.

The Impact of Group Member Familiarity. We know that knowledge of another person's attitudes has been correlated with better relationship functioning in the real world (Sanbonmatsu et al., 2010), as well as improved team performance and task accuracy in VR (Moore & Geuss, 2020). Additionally, a higher degree of familiarity results in a higher level of collaboration in an augmented reality educational environment (Ahmed & Lataifeh, 2023). Tütüncü et al. (2025) found that groups of friends reported higher copresence, showed more positive sentiment, and engaged more frequently in behaviors such as dyadic turn-taking than groups of strangers in virtual reality discussions, and suggested that a moderator plays a critical role in overcoming a lack of familiarity among groups of strangers, recommending 'ice-breaking' activities before moving onto conflict resolution activities, as we do in this study.

In dyads using a CAVE immersive virtual reality setup, we know that friends are more likely to quickly establish methods for tasks and engage in unrelated small talk, unlike strangers (Moustafa & Steed, 2018). Additionally, acquaintances have been found to quickly establish good interpersonal trust among themselves and have better task performance in Collaborative Virtual Environments than strangers (Li et al., 2024), and adding new members to a team results in more closeness to newcomers when the team works in VR than when it operates face-to-face (Fernandez-Espinosa et al., 2025). These findings suggest that familiarity impacts both task performance and entitativity in VR.

Spatiality, Copresence, and Group Cohesion

Establishing common ground and social presence among interactants is central to computer-mediated communication (Clark & Brennan, 1991). In traditional two-dimensional video conferencing, disparities in member influence and the lack of shared spatial coordinates often hinder the formation of common ground, and thereby, team performance and natural communication patterns (Kim et al., 2025). In contrast, co-presence in a shared virtual space allows for co-orientation toward objects and smoother coordination (Meyer & Jucker, 2022). Spatial interfaces for teleconferencing enable important nonverbal communication methods

missing from traditional videoconferencing such as eye contact, body language, and spatially referential gestures like pointing (Neate et al., 2022) as well as additional verbal communication methods including asides and parallel conversations (Sellen et al., 1992), all of which contribute to establishing presence and common ground for groups.

The transition to spatial interfaces, where avatars are volumetrically situated within a user's environment, fundamentally alters individual and group level social dynamics and behaviors. Chen et al. (2025) investigated how the spatial configuration of two-dimensional video chats impacts user interaction and perception by comparing a standard grid-style "Gallery" interface with a spatial, scene-based "Room" interface. Researchers assigned 34 participants to either a collaborative Social Support task, which utilized a circular seating arrangement in the Room setting, or a Topic Discussion task, which placed opposing teams on opposite sides of a virtual classroom. The Room configuration provided visual cues that aided participants in locating team members, orienting allegiances, and using directional body language for turn-taking. Additionally, spatial arrangements shaped social dynamics, as the circular layout fostered group inclusivity while the divided layout enhanced team awareness, albeit sometimes becoming overly confrontational.

In exploring spatial configurations for remote collaboration, Tang et al. (2023) evaluated "Perspectives," a hybrid meeting prototype that digitally composites in-room and remote users into a shared virtual environment, providing each participant with a spatially consistent, unique first-person viewpoint. The researchers conducted a within-subjects study with 60 participants, configured in groups of five (two local, three remote), to compare the spatialized Perspectives mode against standard teleconferencing layouts. Through group discussion tasks and cognitive load assessments, they found that Perspectives was the most preferred interface and significantly reduced cognitive effort, and that spatial layouts broadly improved perceived co-presence over the traditional grid view. Spatial audio also played to natural conversation and reduces cognitive load by making overlapping speech easier to interpret.

To address the structural asymmetry between immersive and traditional platforms in hybrid meetings, Qiu et al (2023) developed ViGather, a teleconferencing system that integrates MR and two-dimensional device users into a shared virtual environment by representing all participants as embodied avatars. For screen-based users, the system leverages webcam feeds to reconstruct three-dimensional upper-body poses and head orientations, allowing them to exhibit spatial non-verbal cues such as gaze, directional pointing, and body turning. The researchers conducted an evaluation with groups of four (two in VR, two on flat devices) and compared ViGather against Meta Horizon, where screen users are typically displayed to those in VR as video feeds on a flat virtual display. The findings highlight the role of unified spatial configurations, as screen-device users in ViGather reported a significantly higher sense of spatial, physical, and self-presence from being spatially represented. Participants noted that providing everyone with a coherent spatial position and a first-person perspective eliminated the

isolating feeling of being in "two separate meetings," demonstrating that shared spatial layouts and equivalent avatar embodiment are essential for co-presence.

In the past, it has been particularly difficult to isolate and vary spatiality without introducing a number of confounding factors such as representation or immersion. In mixed-device teleconferencing, VR users are represented by avatars, while two-dimensional device users are most often represented by webcam streams or profile pictures. When a spatial representation does exist for all users (as in ViGather), VR users are immersed in the conversational space, while two-dimensional device users are not. In two-dimensional teleconferencing, configurations that create a sense of shared spatiality (like Together Mode or Perspectives) do so relative to the interface, but not physical reality; for example, a user may be able to reference that their coworker is to the left of themself on the screen, but is not actually able to “turn to face them”, unlike in VR. In this study, all participants used virtual reality devices, ensuring that all interlocutors were equally immersed and represented, and that the spatiality created was not just within the interface, but the real world.

Avatar Realism and Fidelity

Unlike in past research investigating photorealism and personalization (Salagean et al., 2023), this study allowed participants to embody highly personalized, highly photorealistic avatars; to observe other such avatars via realtime social interactions; to do so in either virtual or mixed reality; and to do so with isolated degrees of spatiality without confounds of immersion or representation. The photorealism, fully tracked facial features and hands, and high resemblance of these avatars to their users are each individually consequential elements, as is the interplay between them, on individual and group dynamics.

Research has often explored how varying graphical realism impacts those who embody and interact with avatars, particularly regarding the “uncanny valley” effect (Mori, 1970). The realism of avatars used in social VR settings has an impact on both the user’s sense of virtual body ownership and the observer's self-perception (Latoschik et al., 2017). Across the body of research, higher fidelity avatars yield more positive social interactions, higher trust, and greater affinity than cartoon-style or absent avatars, a preference that is strengthened in VR (Lee & Ji, 2024; Seymour et al., 2021; Tao et al., 2025; Yee et al., 2007). Avatars based on point-cloud reconstructions increase perceived co-presence, social presence, behavioral impression, and humanness compared to standard virtual avatars (Yu et al., 2021), and groups using realistic avatars in tasks often outperform those using mixed- and cartoon-only avatars (Izzouzi et al., 2024).

An avatar’s level of behavioral realism, the accurate representation of a user’s bodily motions, facial expression, and gaze, can be just as important as its visual fidelity. Limiting an avatar’s nonverbal communication capabilities reduces nonverbal behavior and harms interpersonal outcomes (Herrera et al, 2020), realistic body movements can have subconscious influences on behavior such as increased interpersonal distance (Kimmel et al., 2024), and full

motion-capture avatars can rival or even exceed face-to-face interactions in certain contexts (Rogers et al., 2022).

There is a distinct interplay between these factors: mismatched visual and behavioral realism degrades co-presence (Bailenson et al., 2005; Garau et al., 2003), as does a mismatch in fidelity between an avatar and its surrounding virtual environment (Jo et al., 2017). To achieve the highest levels of presence, embodiment, and trust in interactions, an avatar must combine high visual fidelity, matching behavioral fidelity, and a strong personal resemblance to the user (Izzouzi et al., 2024; Panda et al., 2022; Waltemate et al., 2018), as we do in this study.

Mixed Reality

The impacts of spatiality and avatar realism are well-documented in full VR, where the physical world is entirely occluded. However, modern MR technology allows for high-fidelity video passthrough, situating avatars within the user's perceived physical environment and creating a hybrid spatial ecology. While avatars are commonly studied in VR research, the more limited use of avatars in MR systems (Schäfer et al., 2022) has resulted in, to our knowledge, comparatively little being known about user embodiment of avatars in applications with MR backgrounds (Genay et al., 2022). Consequently, it remains unclear how social presence, cognitive load, and entitativity differ when interacting against a physical passthrough background versus a virtual environment. While users exhibit higher co-presence when interlocutors have real-environment video backgrounds rather than virtually constructed ones (Jo et al., 2017), a direct comparison of live passthrough and a virtual environment as the backdrop of a call with avatars was previously unstudied.

The modality of an interaction, whether a common virtual environment or individual physical passthrough, serves as a perceptual boundary. The mutual assumptions that interactants use to develop shared understanding are impacted by the commonality of their environment, and discrepancies in spatial references during social interactions results in worse synchrony and cognitive load (Han et al., 2026), creating a need for new methods of perspective alignment (Gottsacker et al., 2025). We investigate MR in this study as a means to investigate the impacts on the individual and group level of a teleconferencing environment’s virtuality (virtual or real) and consequent commonality (shared or disparate, respectively), while avoiding the typically confounding lack of visual realism in VR.

Method

Participants

Participants were recruited from a medium-sized private university via the department pool, email, flyers, word-of-mouth, and in-person recruitment and were paid $25 or given research credit. Based on a pre-registered power analysis, we aimed for 132 participants and recruited 143 participants. Two complete four person groups that experienced network connectivity problems resulting in mixed conditions were removed (n=8), as were participants who failed manipulation checks (n=2), resulting in a final baseline n = 133 (24 participants in 2-person groups, 50[2] participants in 3-person groups, 59[3] participants in 4-person groups, all-male = 1, all-female = 10, mixed-gender = 33). Participants (female = 84, male = 45, non-binary or third-gender = 3, transgender = 1) were between 18 and 55 years old (M = 23.25, SD = 6.90) and identified as Asian or Asian-American (n = 59), White (n =34), Hispanic, Latino, or Spanish origin (n = 6), multiethnic (n =21), African, African-American, or Black (n =9), Middle Eastern or North African (n = 1), or preferred not to disclose (n = 3). Participants had varying levels of familiarity with VR (Not at all familiar = 23, A little familiar = 71, Somewhat familiar = 30, and Very familiar = 8) and the Apple Vision Pro itself, as well as with their other group members. No participant had any form of stereo blindness.

Equipment and Setup

Each participant wore an Apple Vision Pro headset (standalone HMD; M2 and R1 chips, visionOS 26.1 or later[4], 3660 × 3200 pixels per eye, 600-650 g, ~100° horizontal FOV, ~90° vertical FOV, 6.5 stereo megapixel color passthrough, six-degree-of-freedom inside-out head, hand, and eye tracking). To standardize the setup process across a large participant pool, all headsets were equipped with generically sized light seals, light seal cushions, and headbands, rather than ones specifically tailored to the participant's head and face. This may have introduced minor variations in physical comfort and light leakage across the sample.

Each headset was connected to an iPad Pro to perform View Mirroring from a Vision Pro via AirPlay, in order to record the headset user's point of view (POV). An additional iPad Pro was used as an external camera to capture participants from the front, utilizing its microphone to record their speech. In the physical space, each participant was seated in a wooden chair with a metal table placed to their right (Figure 1).

The teleconferencing interactions were conducted over a local WiFi network.

[2] After the removal of a participant who failed the manipulation checks
[3] After the removal of a participant who failed the manipulation checks
[4] No OS-level changes relevant to this study were made between the introduction of visionOS 26.1 and the end of the study.

Figure 1

The physical setup of the main experiment rooms. Up to four rooms were used for each group.



Task Materials

Participants completed an adapted version of the Desert Survival Problem, a validated consensus-building exercise frequently used in computer-mediated communication research to elicit naturalistic group negotiation (see Appendix). Participants were informed they had crash-landed in the Sonoran Desert and needed to rank salvaged items by their importance for survival. Because participants in the VR conditions could not reference physical documents as are typically used in this task, all instructions were given verbally.

To ensure equitable participation and balance cognitive load across varying group sizes, the total number of salvaged items scaled with the group size (6, 9, or 12 items). Each participant was verbally assigned three specific items to remember and later discuss with the group (e.g., a flashlight, a knife, and a map). Previous work has found that the co-occurrence of social presence and similar indicators such as mental effort and task load is less likely to result from symmetric tasks, wherein each participant has the same role, tools, and amount of information (Osmers et al., 2021). Therefore, this distribution ensured each participant was given the same amount of information to recall and the same responsibility to participate in the discussion.

Experimental Design

We randomly assigned participants to one of four conditions based on two independent variables, *spatiality* (spatial versus grid) and *modality* (VR versus MR), further described below. While the study was originally designed as a 2x2 factorial experiment, logistical constraints regarding participant scheduling necessitated running groups of varying sizes (dyads, triads, and tetrads). To account for this post-hoc variation, we distributed these different group sizes as evenly as possible across the four core conditions, as described in Table 1.

Table 1
Number of participant groups per condition by size.

	Condition			
Group Size	**Spatial, MR**	**Spatial, VR**	**Grid, MR**	**Grid, VR**
2	4	3	3	2
3	3	5	4	5
4	4	4	4	3

All participants first underwent the same procedures to create their photoreal avatars (Personas) after being instructed by the researcher and handed the headset. All participants then underwent the standard headset calibration procedure for their eyes and hands to ensure accurate representation of their gaze via their avatar. The full experimental protocol and script is presented in the Appendix. All groups completed the same consensus task in the group call with identical time constraints. Groups were randomly assigned to one configuration in a 2 (Spatiality: Spatial vs. Grid) by 2 (Modality: Mixed Reality vs. Virtual Reality) between-subjects factorial design.

For the Spatiality manipulation, the visual rendering of the group members was altered.

In the Spatial condition, avatars utilized the spatial Persona feature of the visionOS FaceTime application, being rendered as three-dimensional Personas positioned volumetrically in the user's real space (Figure 2) or the shared virtual space of the call (Figure 4).

Figure 2

POV screenshots from All Perspectives of a Four-Person Spatial, MR Group



In the Grid condition, avatars were constrained within windows, mirroring a traditional video-conferencing grid layout (Figure 3). In this condition, avatars necessarily had virtual environment backgrounds within their windows, regardless of whether the assigned modality was virtual or mixed reality.

Figure 3

POV screenshots from All Perspectives of a Four-Person Grid, MR Group



For the Modality manipulation, the level of immersion was altered. In the Mixed Reality condition, participants viewed their physical surroundings via stereoscopic video passthrough (Figures 2, 3).

In the Virtual Reality condition, the physical room was fully occluded, and the group shared a synchronized, fully immersive native visionOS environment (Figures 4, 5).

Figure 4

POV Screenshots from All Perspectives of a Four-Person Spatial, VR Group (Bora Bora)



Figure 5

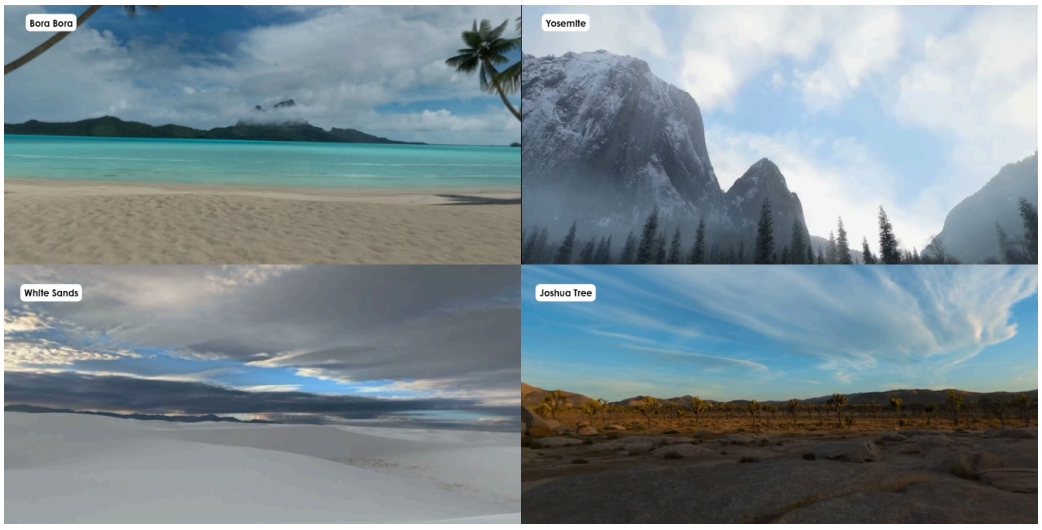
POV Screenshots from All Perspectives of a Four-Person Grid, VR Group (White Sands)



The VR environment was varied per group between native visionOS scenes: White Sands, Yosemite, Joshua Tree, and Bora Bora (Figure 6).

Figure 6

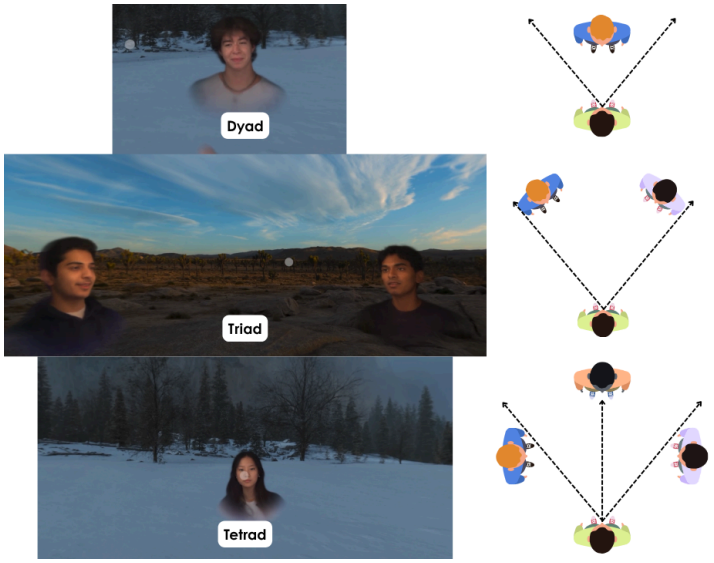
POV Screenshots of Virtual Environments



The placement of spatial avatars or arrangement of avatar windows varied by group size (Figures 7, 8).

Figure 7

POV Screenshots with Bird's-eye View of Spatial Layouts by Group Size



Note: In a four-person spatial group, avatars are spaced such that it is possible to see two, but not all three, other participants simultaneously without moving farther away.

Figure 8

POV Screenshots of Grid Layouts by Group Size



Note: In a two-person Grid call, the aspect ratio of the avatar window is 3:4, whereas in three- or four-person groups, the aspect ratio is 1:1.

Procedure

Upon arrival, participants provided informed consent and completed a pre-study questionnaire on an iPad. Researchers then assisted each participant in generating a photorealistic avatar (Persona) using the Apple Vision Pro's scanning process, and calibrated the headset's native eye- and hand-tracking systems.

To create the conditions of real-world teleconferencing and remove the possibility of visual or audio doubling, participants were then escorted to separate physical rooms and seated in a standardized configuration. Researchers initiated headset point-of-view recording via AirPlay to an iPad Pro and started a secondary external camera recording to capture physical movements and speech. Each participant was instructed on how to join the call once it was ready; due to the sequential nature of equipment setup and troubleshooting across multiple rooms, the wait time between a participant putting on the headset and joining the call varied from approximately 1 to 10 minutes. Once all participants were isolated, a researcher initiated a group FaceTime call, sending a UI window to the headsets prompting each participant to join. As participants joined, a researcher resolved any difficulties and verbally instructed them to engage the grid configuration and fully immersive VR environment if appropriate for their assigned condition. A brief visual synchronization task (simultaneously opening a clenched fist on a verbal countdown) was conducted to align the video recordings for subsequent analysis.

To familiarize participants with the interface and establish a communication baseline, the researcher facilitated a structured icebreaker. Participants sequentially answered two personal questions each (following Bailenson et. al, 2006), followed by a three-minute unmoderated free-conversation period during which the researcher was not in the call.

The researcher then rejoined the call to administer the collaborative task. The Desert Survival scenario was read aloud, and each participant was verbally assigned three specific items to remember and manage. Groups were given exactly 13 minutes to negotiate and agree on a unanimous ranking of all pooled items. Participants were told to refrain from opening other applications, leaving their seats, or removing the headset, and were told to remain on the call if they reached consensus early. The researcher then exited the call. After 13 minutes, the researcher returned to the call and halted the task. Participants were instructed to remove their headsets, completed a post-study questionnaire, and were debriefed and compensated.

Measures

All survey responses were collected digitally using an iPad and Qualtrics software. The pre-study questionnaire captured baseline demographics (age, gender, ethnicity) and participants' prior familiarity with virtual reality headsets, the Apple Vision Pro specifically, and their other group members. To account for potential individual differences in spatial processing and VR tolerance, participants also reported their handedness, participation in spatial-heavy activities

(specific sports and three-dimensional video game genres), and historical susceptibility to motion sickness across various vehicles and activities.

The post-study questionnaire assessed the effects of the experimental manipulations alongside qualitative group dynamics.

Familiarity. In the pre-study questionnaire, participants were asked to describe their familiarity with the other members of their group before the day of the study. These responses were coded into bins of 0 (no familiarity with any other members), 1 (familiarity with one other member), 2 (familiarity with two other members), and 3 (familiarity with three other members). Familiarity was designated whenever a participant indicated they knew they had interacted with another before, or stated their relationship to them (classmate, significant other, roommate, etc.). Total familiarity was expressed as a function of group size.

Manipulation Checks. To verify the efficacy of the experimental conditions, participants were presented with multiple-choice questions accompanied by reference images. They were asked to identify the modality they experienced (the real physical room vs. various virtual environments) and the configuration of the other avatars (spatial vs. grid).

Copresence. Participants rated their perceived co-presence with the group on an 11-item, 4-point scale (1 = never to 4 = always). Items assessed mutual awareness and the illusion of physical proximity (e.g., "I often felt as if my group and I were in the same room together," "I paid close attention to my group"; Cronbach's alpha = [.79]) (adapted from Harms and Biocca, 2004).

Entitativity. Entitativity was evaluated on a 3-item, 4-point scale (0 = never to 3 = always) (e.g., "I felt 'in tune' with my group members," "I felt part of a group") (Cronbach's alpha = [.81]) (Han et al., 2023).

Ostracization. Perceived ostracization was evaluated on an adapted 2-item, 4-point scale (0 = never to 3 = always) (e.g., "I felt left out," "I felt isolated from others") (Spearman-Brown = [0.83]) (Hughes et al., 2004).

Closeness. Participants completed an adapted visual Inclusion of Other in the Self (IOS) scale, selecting from a series of increasingly overlapping circles ranging from 1-7 to represent their relationship with the group (adapted from Aron et al., 1992).

Task Load. The physical, mental, and emotional toll of the communication setup was assessed on a 3-item, 4-point scale (0 = not at all to 3 = a lot). Participants rated the physical and mental demand of the group call, and feelings of frustration (adapted from Hart & Staveland, 1988).Due to low Cronbach's alpha for the items adapted from the original scale (Cronbach's

alpha = 0.60), we investigated each item, namely: (1) physical load, (2) emotional load, and (3) frustration, separately.

Comfort. The level of physical and emotional comfort participants felt was assessed on a 2-item, 4 point scale (0 = not at all to 3 = a lot) (adapted from Hart & Staveland, 1988). Due to low Spearman-Brown correlation (r = 0.53), we looked at the two items separately.

Post-Collaboration Reflection. Finally, participants provided open-ended text responses describing the strategies their team used to reach a consensus, the emergence of specific roles (e.g., leader or mediator), and the interpersonal dynamics they would seek to replicate in future collaborative tasks.[5] From the open-ended responses, we extracted four outcome variables, three of which used Linguistic Inquiry and Word Count (LIWC) (Tausczik & Pennebaker, 2010) to quantify linguistic patterns and another used a manual coding scheme for identifying the presence of role division.

We extracted three LIWC categories for participants' open-ended responses after removing responses shorter than 10 words, which resulted in a sample size of 132. We reference past VR literature in picking the 10-word threshold on studying linguistic patterns using LIWC (DeVeaux et al., 2024). We first extracted *word count* as a measure of effort in completing the task (M=43.51, SD=26.61). We then examined the categories of *first-person plural pronouns* (M=6.65, SD=4.30) and *affiliation* (M=10.67, SD=9.85), which were calculated by extracting the percentage of words in the written response that matched a certain linguistic category (e.g., "we" for first-person plural pronoun).

The final measure is the presence of the labor division, as reflected through the participant's written response. One member of the research team first coded a participant's responses, with those mentioning the presence of division of labor or role assignment being coded as 1, and 0 otherwise. Another coder then double-checked this code, verifying the correctness of the initial labels. There were no disagreements on the labels (N_0=105, N_1=28).

Face Attention. To quantify participants' visual attention toward their collaborators, we analyzed the screen recordings of their point-of-view from within the headset. This continuous variable serves as a spatial proxy for interpersonal gaze, reflecting the frequency and consistency with which a participant maintained their collaborators within their active field of view throughout the task. We extracted face detection metrics using a custom computer vision pipeline built on the InsightFace library, utilizing the buffalo_l model. To optimize computational efficiency and process the recordings at scale, the model was restricted strictly to spatial bounding box detection.

[5] The responses were collected across two questions, which we binned for linguistic analysis.

Because the POV screen recordings exhibited variable frame rates (VFR) and occasionally dropped frames due to localized network latency or rendering load, relying on standard index-based frame counting would result in severe temporal desynchronization. To account for this, we implemented a timestamp-anchored extraction method. The algorithm advanced through the video file and queried the playhead's absolute millisecond timestamp, decoding a visual frame only at strict 500-millisecond intervals. This ensured a reliable, standardized sampling rate of exactly two frames per second across all recordings.

For each sampled frame, the pipeline recorded the absolute count of detected faces. From this frame-level data, we extracted three distinct proportional metrics for our binomial modeling: zero-face attention (the proportion of frames with no visible faces, representing complete visual disconnection across all conditions), single-target attention (the proportion of one-face frames, representing deliberately isolated focus within Spatial layouts), and intentional disengagement (the proportion of frames containing fewer than the functionally available maximum number of faces, representing deliberate looking away within Grid layouts).

Data Analysis

Self-report data analysis was run in R using dplyr and lmer4 packages. Incomplete data were treated as missing at random, and statistical significance was evaluated at $\alpha = 0.05$. Individual differences in how individuals in groups self-reported their entitativity, ostracism, co-presence, cognitive load, and comfort were examined using linear mixed effects models where the spatiality and modality, in addition to group size and group familiarity, were treated as fixed effects and group was treated as a random effect. When a mixed effects model resulted in a singular fit, meaning that the random effect of group did not account for considerable variance in the outcome variable, we built a linear model with only fixed effects. For predicting the presence of division of labor, we built mixed-effects logistic regressions with the same set of fixed and random effects used for the linear models.

To address visual artifacts introduced by the visionOS interface, specifically the transient ‘Mini-Self View’ native to the Grid configuration, we implemented an algorithmic filtration pipeline prior to statistical modeling. Detected faces were removed from the dataset if their pixel area fell below a baseline threshold and they were simultaneously less than 30% of the average size of the other visible faces in the frame. While this successfully removed the majority of self-view instances, the algorithmic filtration was not perfect, leaving a known marginal error rate of uncaught self-view frames in the final dataset.

POV face-tracking data analysis was conducted in R using the lme4 package. Statistical significance was evaluated at $\alpha = 0.05$. Visual attention behaviors were examined using binomial Generalized Linear Mixed Models (GLMMs). Spatiality, modality (MR/VR), group size, and familiarity were treated as categorical fixed effects. To account for the non-independence of participants interacting within the same session, group ID was included as a random intercept in

all models. To inherently control for varying trial durations across participants, the models predicted the weighted proportion of target frames out of the total analyzed frames. Finally, to prevent structural zeros from invalidating the models, the dataset was subset to test specific behavioral choices: single-target attention was modeled exclusively for groups of three and four within the Spatial layout, and intentional disengagement (viewing fewer than the maximum available faces) was modeled exclusively within the Grid layout.

Results

Preliminary Analyses and Manipulation Checks

Participants were excluded due to technical issues (n=8) or failing the manipulation checks (n=2), resulting in a baseline n = 133 for survey-based measures.

Table 2

Means and Standard Deviations of Survey-Derived Variables by Spatiality and Modality

	Means (Standard Deviations)			
Variable	**Spatial and MR**	**Spatial and VR**	**Grid and MR**	**Grid and VR**
Co-Presence	2.93 (0.37)	2.95 (0.37)	2.94 (0.43)	2.96 (0.34)
Entitativity	3.22 (0.66)	3.19 (0.58)	3.41 (0.60)	3.43 (0.10)
Ostracism	1.05 (0.19)	1.15 (0.33)	1.14 (0.36)	1.10 (0.27)
Closeness	4.33 (1.80)	4.50 (1.18)	4.58 (1.52)	4.97 (1.33)
Physical Demand	0.79 (0.96)	0.75 (0.87)	0.70 (0.64)	0.74 (0.89)
Emotional Demand	1.30 (0.95)	1.14 (0.80)	1.21 (1.05)	1.16 (1.00)
Physical Comfort	1.67 (0.85)	1.92 (0.84)	1.79 (0.65)	1.97 (0.84)
Emotional Comfort	2.00 (0.75)	2.33 (0.63)	2.33 (0.74)	2.32 (0.75)
Frustration	0.27 (0.52)	0.33 (0.53)	0.24 (0.50)	0.32 (0.65)

Group-level Associations

Co-Presence. Linear mixed effects models revealed no significant differences were present for spatiality (b = 0.01, SE = 0.08, df = 40.33, p = 0.89) and modality (b = 0.02, SE = 0.08, df = 40.49, p = 0.84) on co-presence. No significant effects were found for group size (b = -0.07, SE = 0.05, df = 49.13, p = 0.22) and prior familiarity (b = 0.001, SE = 0.07, df = 116.26, p = 0.99).

Entitativity. Linear mixed effects models revealed marginal significant differences were present for spatiality (b = 0.22, SE = 0.12, df = 39.76, p = 0.07) and none for modality (b = -0.02, SE = 0.12, df = 39.93, p = 0.85). Significant effects were found for group size (b = -0.21, SE = 0.08, df = 48.76, p = 0.01). Participants in groups that had prior familiarity with other members of the group had marginally higher entitativity compared to those that did not (b = 0.20, SE = 0.11, df = 114.68, p = 0.07).

Ostracism. Linear mixed effects models revealed no significant differences were present for spatiality (b = 0.02, SE = 0.05, df = 46.39, p = 0.76) and modality (b = 0.04, SE = 0.05, df = 46.80, p = 0.52) on ostracism. Significant effects were found for group size (b = 0.08, SE = 0.04, df = 59.41, p = 0.049) where higher group sizes had higher ostracism compared to smaller groups. No significant effects were found for familiarity (b = -0.002, SE = 0.06, df = 103.39, p = 0.97).

Closeness. Linear mixed effects models revealed no significant differences were present for spatiality (b = 0.36, SE = 0.31, df = 33.98, p = 0.25) and modality (b = 0.25, SE = 0.31, df = 34.14, p = 0.43) on closeness. No significant effects were found for group size (b = -0.23, SE = 0.21, df = 42.26, p = 0.29) where higher group sizes had higher ostracism compared to smaller groups. No significant effects were found for familiarity (b = 0.23, SE = 0.29, df = 112.08, p = 0.42).

Task Load and Discomfort

Linear and linear mixed-effects models revealed no significant main effects of spatiality, modality, group size, or familiarity on physical demand, mental demand, frustration, physical comfort, or emotional comfort.

Table 3

Full Statistical Details for Task Load and Discomfort

Dependent Variable	Predictor	b	SE	df	p
Physical Demand	Spatiality	-0.05	0.15	128	0.74
	Modality	-0.01	0.14		0.99
	Group Size	0.13	0.10		0.23
	Familiarity	0.04	0.16		0.81
Mental Demand	Spatiality	-0.04	0.16	128	0.82
	Modality	-0.11	0.16		0.52
	Group Size	0.17	0.12		0.16
	Familiarity	-0.08	0.18		0.16
Frustration	Spatiality	-0.02	0.10	38.37	0.88
	Modality	0.06	0.10	38.69	0.53
	Group Size	0.10	0.07	49.79	0.17
	Familiarity	0.15	0.10	100.52	0.14
Physical Comfort	Spatiality	0.09	0.15	39.44	0.56
	Modality	0.23	0.15	39.84	0.12
	Group Size	0.03	0.10	51.98	0.72
	Familiarity	-0.18	0.15	97.27	0.22
Emotional Comfort	Spatiality	0.18	0.14	39.47	0.22
	Modality	0.17	0.15	39.66	0.25
	Group Size	-0.17	0.10	48.72	0.10
	Familiarity	0.14	0.14	113.03	0.33

Observations of Facial Attention

Table 4

Means and Standard Deviations of Point-of-View Recording Analyses

Group Size	Spatiality	Modality	0-Face Mean	0-Face SD	1-Face Mean	1-Face SD	2-Face Mean	2-Face SD	3-Face Mean	3-Face SD	4-Face Mean	4-Face SD
2	Spatial	MR	3.57%	4.15%	96.31%	4.18%	0.12%	0.16%	0.00%	0.00%	0.00%	0.00%
2	Spatial	VR	7.26%	7.33%	92.73%	7.32%	0.01%	0.03%	0.00%	0.00%	0.00%	0.00%
2	Grid	MR	0.25%	0.54%	99.75%	0.54%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
2	Grid	VR	4.09%	2.89%	95.91%	2.89%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
3	Spatial	MR	1.39%	1.92%	58.66%	21.60%	39.92%	22.58%	0.02%	0.05%	0.00%	0.00%
3	Spatial	VR	5.99%	4.23%	49.14%	16.20%	44.87%	15.89%	0.00%	0.00%	0.00%	0.00%
3	Grid	MR	1.68%	3.56%	1.76%	3.38%	95.97%	8.09%	0.59%	1.32%	0.00%	0.00%
3	Grid	VR	3.10%	3.23%	4.58%	3.97%	90.83%	8.15%	1.49%	2.85%	0.00%	0.00%
4	Spatial	MR	2.43%	4.73%	22.80%	13.74%	73.56%	14.57%	1.21%	4.40%	0.00%	0.02%
4	Spatial	VR	2.58%	3.78%	25.27%	16.16%	72.11%	16.17%	0.04%	0.07%	0.00%	0.00%
4	Grid	MR	2.23%	4.28%	1.79%	2.71%	4.82%	5.36%	91.03%	11.18%	0.14%	0.38%
4	Grid	VR	2.11%	2.79%	2.76%	4.46%	6.69%	8.86%	88.43%	13.45%	0.00%	0.00%

Note: This table shows the percentage of time a given number of faces were present, restricted to the task portion of the call, and after algorithmic removal of self-view[6]

Of the baseline n = 133 participants, some were excluded specifically from face attention analysis due to failure of the point-of-view recording (n=5). Therefore, for face attention measures, we evaluate the data from n = 128 participants.

To evaluate visual attention without violating model assumptions due to structural zeros, we modeled three distinct behavioral metrics using binomial generalized linear mixed models (GLMMs): zero-face attention, single-target attention, and intentional disengagement.

Across all configurations, group size and familiarity were non-significant in predicting the frequency of participants completely disconnecting from the group (seeing zero faces). However, spatiality and modality significantly impacted complete visual disconnection. Participants in the Grid condition saw zero faces significantly less often compared to those in the

[6] The visionOS FaceTime application features two distinct methods for a user to view their own Persona during a call; a persistent window the size of a standard avatar window (Persona Mirror), and a transient window about 1/3rd of the size of a regular grid window (Self-View), which appears when selecting interlocutors in the Grid condition. The Self-View is only visible within the grid configuration, whereas the Persona Mirror is accessible in both the Spatial and Grid configurations. While all participants were instructed not to alter the configuration of the call or open any additional windows, researchers did not intervene if they did so. While detected Self-View faces were algorithmically removed, Persona Mirror faces could not be algorithmically distinguished from actual faces, and were therefore left intact due to their insignificant occurrence.

Spatial condition (b = -1.88, SE = 0.24, z = -7.87, p < .001). Conversely, participants in the VR modality saw zero faces significantly more often compared to those in the MR modality (b = 0.86, SE = 0.24, z = 3.60, p < .001).

To analyze genuine social choice regarding focus, we isolated the Spatial condition for groups of three and four to predict the frequency of single-target attention (the proportion of time spent focusing on only one face when multiple are available). Modality type was non-significant (b = -0.11, SE = 0.15, z = -0.77, p = .44). Participants in groups of three exhibited significantly higher single-target attention compared to those in groups of four (b = 1.08, SE = 0.14, z = 7.44, p < .001). Additionally, unfamiliar groups showed a marginally higher frequency of single-target attention compared to familiar groups (b = 0.03, SE = 0.02, z = 1.74, p = .082).

Finally, we evaluated intentional disengagement exclusively within the Grid condition by predicting how often participants actively looked at fewer than the functionally available maximum number of faces. Participants in the VR modality intentionally disengaged significantly more often than those in MR (b = 1.22, SE = 0.42, z = 2.91, p = .004). Group size also significantly impacted this behavior; compared to groups of four, participants disengaged significantly less often in groups of three (b = -1.50, SE = 0.48, z = -3.13, p = .002) and pairs (b = -2.69, SE = 0.56, z = -4.81, p < .001). Unfamiliar groups exhibited significantly less intentional disengagement compared to familiar groups (b = -0.19, SE = 0.03, z = -6.70, p < .001).

Group Dynamics through Open-Ended Responses

Spatiality was non-significant in predicting word count (b=-4.27, SE=4.52, p=.35), and participants in the VR condition wrote longer responses compared to those in the MR condition (b=11.83, SE=4.52, p=.010). There were non-significant effects for group size (b=0.94, SE=3.15, p=.76) and group familiarity (b=0.38, SE=4.79, p=.94). The use of the first-person plural pronoun was non-significantly predicted by spatiality (b=1.19, SE=0.82, p=.16) and modality (b=-0.86, SE=0.82. p=.30). It was not significantly predicted by group familiarity (b=0.63, SE=0.78, p=.42), and participants in bigger groups tended to use fewer first-person plural pronouns compared to those in smaller groups (b=-1.66, SE=0.56, p=.0044). Similar trends were found for affiliation, namely, that it was non-significantly predicted by spatiality (b=1.71, SE=1.20, p=.16) and modality (b=-1.22, SE=1.20, p=.31). Group familiarity did not significantly predict affiliation (b=1.16, SE=1.07, p=.28), and participants in larger groups again were predicted to use fewer affiliation words than those in smaller groups (b=-1.94, SE=0.81, p=.020).

Finally, for predicting the presence of division of labor, we found no main effects in spatiality (b=-0.27, SE=0.56, p=.63) and modality (b=-0.064, SE=0.57, p=.91). Group familiarity was non-significant in predicting the presence of division of labor (b=-0.51, SE=0.55, p=.36), and those in larger groups were predicted to be more likely to report a division of labor (b=1.43, SE=0.50, p=.0043).

Discussion

Summary of Results

This study investigated how spatiality (spatially located vs. grid) and modality (VR vs. MR) shape group cohesion, visual attention, and communication dynamics during a collaborative consensus task. The results contrast expectations of spatial computing and actual subjective outcomes. Most notably, the architectural affordances of the teleconferencing system had almost no impact on participants' subjective experiences of task load, comfort, or co-presence. Instead, structural group dynamics, and specifically group size, emerged as the dominant force shaping social outcomes.

The anticipated social benefits of physical-world spatial semantics did not translate into greater group cohesion. As group size scaled from dyads to four-person teams, participants instead reported significantly lower entitativity ($p = 0.01$) and higher feelings of ostracism ($p = 0.049$), regardless of whether they were interacting via spatial avatars or a grid layout. This effect was explicitly mirrored at the linguistic level. When reflecting on their team's consensus strategies, participants in larger groups used significantly fewer inclusive first-person plural pronouns ("we") ($p = 0.0044$) and affiliation words ($p = 0.020$). Conversely, these larger groups relied more heavily on references to rigid division of labor ($p = 0.0043$). This structural effect suggests that while spatial interfaces strive to recreate the intimacy of physical presence, they cannot inherently bypass the difficulties introduced by scaling up a collaborative task.

Contrary to Hypothesis 1, neither the spatial representation of interlocutors nor the modality significantly mitigated the cognitive or physical toll of the interaction. Linear models revealed no significant differences across any dimension of task load, including physical demand, mental demand, frustration, or comfort. While previous literature suggests spatial configurations can decrease cognitive load compared to standard teleconferencing in traditional screens, the current findings indicate that imposing these spatial layouts in headset-based MR and VR environments does not yield the same restorative benefits.

While subjective load remained unaffected, the system manipulations did marginally impact group identity. Spatial groups reported marginally lower entitativity than traditional grid groups ($p = 0.07$). While this effect is marginal and should be interpreted with caution, it challenges the assumption that spatialization increases group cohesion. The introduction of photorealistic avatars into a user's peripersonal space may trigger avoidance behaviors or highlight the uncanny nature of the mediation, disrupting the safe, established social boundary that standard grid views provide. Notably, baseline familiarity mitigated some of this friction, with previously acquainted groups trending toward higher entitativity ($p = 0.07$).

Though subjective social outcomes were primarily dictated by group size, the behavioral and attentional mechanics of the interactions were heavily dictated by the spatial layout (Grid vs. Spatial) and the modality (VR vs. MR). This produced a paradox regarding shared visual

context. In the fully immersive VR condition, all participants shared the exact same synthetic environment, providing a common ground for spatial reference. Conversely, participants in the MR condition were visually isolated, each viewing their own disparate physical room via passthrough. Theoretically, the shared reality of VR should foster greater visual attention on collaborators.

However, the point-of-view face-tracking analysis revealed the opposite. Participants in VR spent significantly more time completely visually disconnected from their group (seeing zero faces) compared to those in MR ($p < .001$). The structural layout also altered overall engagement: the Grid interface significantly decreased the time spent seeing zero faces compared to the dispersed Spatial layout ($p < .001$). Even within the Grid layout, where the interface clusters faces together to make full-group viewing physically effortless, participants in VR intentionally disengaged (viewing fewer than the maximum available faces) significantly more often than those in MR ($p < .01$). When using the Spatial layout, single-target attention was driven by group size rather than the modality, with groups of three isolating their focus onto a single person significantly more often than groups of four ($p < .001$).

The data suggest that the shared virtual environments in VR acted as an environmental distractor. This visual distraction in VR was accompanied by differences in post-task reflection. Participants in immersive VR generated significantly longer open-ended responses than those in MR ($p = .010$). Because VR groups spent less time engaged in direct visual attention, critical nonverbal communication channels may have been compromised. Consequently, these groups may have needed to overcompensate with explicit verbal negotiation to reach consensus, resulting in longer post-hoc explanations of their strategy.

Evaluating the complete collaborative process requires further integration of objective interaction metrics. Subsequent analyses of conversational content, precise gaze-pointer tracking, and overt multitasking behaviors will be necessary to determine whether the environmental distraction observed in VR, or the presence of spatiality, ultimately translated into noticeable collaborative disengagement.

Implications

For Theory. The findings challenge current theoretical assumptions regarding behavioral and social dynamics in computer-mediated communication. Existing literature frequently posits that shared spatial coordinates and increased avatar fidelity fundamentally improve group function, natural communication patterns, and reduce cognitive load. However, our results demonstrate that the structural affordances of spatiality and environment virtuality do not inherently supersede the friction introduced by scaling group size. The finding that larger groups reported significantly lower entitativity ($p = .01$) and higher ostracism ($p = .049$) regardless of whether they used spatial avatars or a grid layout suggests that theories of computer-mediated communication must examine group structure as well as mediated spatiality.

The results also introduce a theoretical paradox concerning environmental backdrop and the establishment of common ground. Foundational theories suggest that co-presence in a shared virtual space allows for co-orientation and smoother coordination. Yet, participants in the fully immersive, shared VR environment exhibited significantly lower rates of visual engagement with interlocutors compared to those in non-shared MR environments. Specifically, participants in the VR condition completely visually disconnected (seeing zero faces) significantly more often than those in the MR condition (Est = 0.86, SE = 0.24, $z = 3.60$, $p < .001$). Even within the Grid interface where full-group viewing is physically effortless, participants in VR intentionally disengaged from the group significantly more often (Est = 1.22, SE = 0.42, $z = 2.91$, $p < .01$).

This indicates that shared virtual environments may distract rather than ground users. Consequently, theoretical models must account for the cognitive cost of environment processing, which may detract from the social processing necessary for cohesion. This distraction may have forced groups to overcompensate with explicit, complex verbal negotiation, as evidenced by VR groups writing significantly longer post-hoc explanations of their strategy compared to those in MR ($b = 11.83$, SE = 4.52, $p = .010$).

Finally, the marginal finding that Spatial groups reported lower entitativity than traditional Grid groups ($p = .07$) implies that imposing physical-world norms, such as peripersonal space, onto mediated avatars may introduce unintended avoidance behaviors. This requires expanding Expectancy Violations Theory within XR, and investigating whether photorealistic embodiment in close virtual proximity disrupts established social boundaries that two-dimensional grids comfortably enforce.

For Practice. For developers and enterprise adopters of XR teleconferencing, these findings necessitate a critical re-evaluation of design priorities. The current industry trajectory assumes that maximizing spatial realism and environmental immersion results in better collaborative experiences. However, the data reveal that spatial configurations do not automatically mitigate the cognitive or physical toll of remote interactions. In practice, forcing users into three-dimensional spatial arrangements may actually increase social friction compared to standard grid layouts. Teleconferencing platforms should reconsider the push to deprecate traditional two-dimensional grid views, as these structures provide a safe, established social boundary that users may prefer when navigating the complexities of multi-person interactions in less familiar modalities like VR.

Additionally, the environmental distraction observed in fully immersive VR provides concrete, actionable data for meeting facilitation. If the goal of a collaborative session requires high visual attention on team members, practitioners should default to Mixed Reality passthrough rather than shared virtual environments. Rather than maintaining consistent views of their peers, participants in the VR condition completely visually disengaged (seeing zero faces) significantly more often than those in the MR condition. Even when utilizing a Grid interface

designed to allow for viewing the entire group, participants in VR intentionally disengaged (viewing fewer than the maximum available faces) significantly more often than those in MR. The familiar, static nature of a user's physical room naturally funnels visual attention toward the avatars, whereas dynamic VR backgrounds pull gaze away. As our VR conditions utilized Apple's highly detailed, beautiful nature scenes (e.g., Joshua Tree, Bora Bora), including some animated elements such as water or swaying tree branches, they contrasted significantly with the unstimulating, static university study room seen in MR. If VR is the format of a meeting, organizers should take care to choose environments with the least potential to distract. If maximizing simultaneous visual attention is the primary objective, facilitators should pair an MR or mundane virtual environment with a grid-style interface. The Grid layout significantly decreased the frequency of participants looking completely away from the group compared to a dispersed Spatial layout.

Finally, because group size emerged as the primary factor for social outcomes, platform designers must build tools that explicitly support group moderation for larger teams. As participants scaled from dyads to tetrads, they used significantly fewer inclusive first-person plural pronouns and affiliation words, and relied more on references to division of labor. XR systems must integrate structural management tools rather than relying solely on spatial semantics to organize group dynamics.

Limitations of Current Study and Future Directions

While the current study utilized avatars with high visual and behavioral fidelity to minimize the detrimental effects of unrealistic mediation, several limitations must be addressed in future work. First, the computer vision analysis relied on a spatial proxy for interpersonal gaze, establishing metrics for zero-face attention, single-target attention, and intentional disengagement. Because frames were sampled at a strict interval of exactly two frames per second, these metrics capture the frequency of faces within a user's field of view but lack the granularity of precise gaze-tracking. Future work must integrate more precise analysis of the captured gaze data to determine whether the manipulations of spatiality and environment impact the distribution of focal subjects and micro-gaze movements.Evaluating the complete scope of the collaborative process requires incorporating additional objective metrics, such as conversational content analysis, overt multitasking behaviors, and body motion-derived measures of synchrony. The data necessary to support these evaluations were gathered during this study, and are pending analysis for inclusion in the published version of this work.

Second, the study employed a highly structured Desert Survival consensus-building task designed to ensure equitable participation. Because participants were verbally assigned exactly three specific items to remember and discuss, this controlled for asymmetric information, but it may not generalize to inherently asymmetric or naturalistic remote work. Tasks with unequal role distribution, passive spectator meetings, or creative brainstorming might yield different behavioral and psychological outcomes and should be investigated specifically.

Third, the behavioral analysis was restricted exclusively to the 13-minute consensus task. It remains unknown whether visual attention behaviors differed during the structured icebreaker, the unstructured free-conversation period, or the instructional phases. Additionally, participants were often complete strangers before their session. Our behavioral models indicated that familiarity significantly alters visual attention; unfamiliar groups exhibited significantly less intentional disengagement than familiar groups ($p < .001$) and marginally more single-target attention ($p = .082$), suggesting a heightened need to continuously monitor unknown interlocutors. Given that previous dyadic research shows relationship functioning and team performance improve with familiarity over time, future longitudinal work should investigate whether the visual attention patterns of strangers transition into the patterns of friends across repeated interactions.

While our pre-registered power analysis targeted 132 participants in groups of four for a 2x2 factorial design, logistical constraints necessitated running groups of two, three, and four. The sample remained evenly distributed across the four core conditions, maintaining adequate statistical power to test the main effects of spatiality and modality. However, including group size as an independent variable functionally shifted the models to a 2x2x3 design. Because the sample size was not calculated for twelve cells, the study is underpowered for detecting complex interaction effects with group size. Consequently, while the main effects of spatiality and modality are robust, all findings related to group size must be interpreted as exploratory. Future research should utilize samples explicitly powered to test group size as a primary independent variable, given the findings here.

Finally, the unmoderated nature of the session introduced uncontrolled behavioral variance. Despite instructions to the contrary, participants occasionally manipulated the visionOS interface by moving or resizing interlocutor windows or opening persistent Persona mirrors. Future studies must utilize strict software-level controls, currently unsupported on visionOS, or active moderation to prevent this variance. Additionally, because the unmoderated task forced groups to self-organize, and larger groups subsequently struggled with cohesion, future studies should introduce assigned roles to determine if structured leadership can overcome the friction inherent to scaling group sizes in spatial teleconferencing.

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Consent to participate

Written informed consent was obtained from all participants prior to their involvement in the study.

Ethical considerations

The recruitment and experiment process were approved by the university's IRB (ID: 72823).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

Appendix

Experimental Protocol and Script

Part 0: Introduction, Consent, Pre-Questionnaire.

Which participants sit in which of the four chairs is random: they come in as a group and are told to sit in whichever of the chairs they like.
Participants are numbered (ID) Left to Right by chair.
Headsets 1-4 are always given in L-to-R order, and participants are assigned to the room corresponding to their headset.

1. Thank you for participating in our study. First, please read through this consent form and sign, and let me know if you have any questions.
2. Now I'll give each of you an iPad with a brief questionnaire. Let me know if you have any questions.
3. [To each Participant] You are participant [ID].
   Hand each participant an iPad to take the pre-questionnaire.
4. In this study, we want to understand how people communicate with each other using virtual reality and avatar technology. We'll help you each create an avatar using these headsets. You'll then go into separate rooms, and we will help you join a group call using your avatars. There will be some icebreaker questions, and then a collaborative verbal activity. At the end, there will be another questionnaire like the one you just completed. Does anyone have any questions?

Part 1: Persona + Calibration.

5. First, we'll help you create an avatar called a Persona. We will start the process on the headset for you, then hand the headset to you to hold in front of you. The headset will give you instructions out loud, such as "turn your head to the left", "turn your head to the right", and ask you to make facial expressions. At the end it will say "capture complete, put Vision Pro back on", at which point you can hand it to one of us to save the avatar to the headset.

   The Research Personnel put on a headset, open "Settings", scroll to "Persona" and select it, select "Get Started", scan their own hands, press "Continue" on the video, then press "Begin Capture", take off the headset, and hand give it to the participant.
6. If you have any questions or difficulty with this process, please let us know and we can assist you.

   Once it says "Put Vision Pro back on" out loud, Research Personnel put it back on and press "Continue", "Done", and "Save". Then, hand back to the participant to hold.

7. Next, we're going to calibrate the headset for your eyes and hands. First, please notice this circular button, the dial button, which rotates and can press down. We may need to refer to it later.
8. Everyone can put the headset on and adjust the velcro straps, one on top and one on the back. Let us know if you need assistance adjusting the straps.
9. If it says "Realign the displays", you can press and hold the button it shows you until you see a green checkmark.

Once everyone has their headset on:

10. Does everyone see "Enter passcode"? You should reach out in front of you with your index finger and press "1 0 0 0" (one thousand).
11. Is everyone past the passcode screen?

Once everyone is in (eyes showing)

12. We will start the calibration by pressing a few buttons on your headset. You can follow the instructions it gives you; when it shows you colorful circles, you should look at them with your eyes, hold your hand comfortably in front of you, and tap your index finger and thumb together like this (demonstrate) to select them.
13. To engage calibration mode:

Top Button press 4x
Digital Crown press 1x ("No Optical Inserts")
Digital Crown scroll down (clockwise)
Digital Crown press 1x ("Redo Eye & Hand Setup")

14. Once the calibration is finished, you can take off the headset and set it in your lap.

Once everyone has taken off their headset:

15. We're now going to move into different rooms for the next part of the study. [To the room 1 person:] You'll be in this room, so you can sit tight for now.

Lead #4 to **Room 4**. Consult **Subsection**. Return to Room 1.
Lead #3 to **Room 3**. Consult **Subsection**. Return to Room 1.
Lead #2 to **Room 2**. Consult **Subsection**. Return to Room 1.
Consult **Subsection** for participant 1.

Subsection: In-Room Directions + Casting/Recording.

16. You can have a seat in this wooden chair and if you have your phone on you, please place it on the table next to you. I will take the headset to set up the next part.

Open the corresponding iPad and check its name, then open the Vision Pro app.
Put on the headset, realign the display if necessary, and enter the passcode.
Open the **Control Center** and hit **Mirror My View**
Select the **iPad** shown with the correct name.
Select 'Speaker' for audio
Go back to the home screen.
Return the headset to them. Once they sign back in, begin Screen Recording on the Screen Recording iPad, and begin Camera Recording on the body iPad.

17. You can put the headset back on. You may need to re-adjust the velcro and re-align the displays. Once that's done, please enter the passcode to the headset again.
18. Once everyone is set up, we will start the call, and a 'Join' button will appear in front of you. You can press it with your index finger, or look at it and pinch. Please wait for a few minutes while everyone else is being set up; please don't press any buttons, open any apps, take off the headset, or move your chair while you wait.

LEAVE THE ROOM THEY ARE IN AND CLOSE THE DOOR!

Part 2: Call + Icebreakers.

Begin the group FaceTime call by texting 'Begin' to the "Study" thread.
Research joins the call (audio-only).

19. Hi everyone, for this next part, I will be giving instructions over audio. So, please confirm that you can hear me.

IF IT IS A GRID CONDITION:

20. Everyone, please look at someone else in the call and pinch your finger and thumb together as if to select them. A menu bar should appear. Now, please look at the '3 dots' button on the right and pinch to select it. Now, please look at and select "Windowed mode".
21. Now, please look straight ahead of you, and press down and hold the dial button for one or two seconds until your view is re-centered.

IF IT IS A VR CONDITION:

22. If you remember the dial button, the button you press to re-align the displays, please rotate it all the way clockwise until your surroundings are replaced with a virtual environment.

23. For a timing calibration, I am going to count to 5. Everyone, please hold a fist in front of your face. When I say "5", please open your hand all the way. 1, 2, 3, 4, 5.
24. We are now going to ask some icebreaker questions. We're going to go in the order of your participant IDs from earlier, so starting with participant [#].
    a. Where do you live on campus, or do you live off campus? (1)
    b. Where did you grow up? (2)
    c. What are you majoring or did you major in? (3)
    d. Do you have any siblings? (4)

e. What line of work are your parents in? (1)
f. What's the scariest thing that's ever happened to you? (2)
g. What has been the most stressful event of the last six months for you? (3)
h. What's the longest relationship you've ever been in? (4)

25. Thank you for answering those questions. Before we start the collaborative activity, we'd like to give you three minutes of conversation time. Please don't open any apps, change any settings, or leave your chair, but you can talk with each other about whatever you like. So, I will leave the call and I'll be back in 3 minutes.

Exit call. 3 minute timer. Once this is done, move on to Part 5

Part 3: Task (Desert Survival) + Post-Questionnaire.

26. We're going to move on to the collaborative activity. Imagine this scenario:
27. It is approximately 10:00 AM on a mid-August day and you have just crash-landed in the Sonoran Desert of the southwestern United States. The pilot and co-pilot were killed in the crash, and the aircraft is destroyed, but all of you remain uninjured. The pilot was unable to notify anyone of your position before the crash. You are 65 miles (about 100 km) from the nearest clear road or habitation. The immediate area is flat and barren, aside from occasional cacti. The last weather report indicated temperatures would reach 110 degrees F (43 C) today.

    From the wreckage, your group has salvaged 12 items. Your task is to rank these twelve items in order of importance for your survival. Number 1 is the most important; Number 12 is the least important.

    Each of you has 3 of the items and is responsible for remembering to discuss and rank them with your group:

    participant [ID1], you have a flashlight, knife, and map[7]; participant [ID2], you have a compass, a roll of gauze, and a pistol; participant [ID3], you have a parachute, one gallon of water (about 4L), and sunglasses; participant [ID4], you have 2 quarts of vodka (about 2L), four rain jackets, and a cosmetic mirror. Would anyone like me to repeat their items?
28. The group will now have 13 minutes to reach a unanimous consensus. You must agree on the exact ranking of all items. If you finish with this task early, please remain on the call and talk about anything you like. Please do not open any apps, change any call settings, or take the headset off until I rejoin the call in 13 minutes. You may begin discussing now; good luck!

Once 13 minutes have passed:

29. Hi everyone, that is the end of the collaborative activity. The only thing left is the final questionnaire about your experiences today; I will come around and give you the iPad to take the questionnaire. You can now take the headset off and set it on the table next to you.

[7] repeat all items once

Return to their rooms. Stop the screen recording on the Screen Recording iPad, stop the recording on the body iPad, and open the Post Questionnaire on the Screen Recording iPad. Hand to the participant and remind them of their ID.

Bring the Vision Pro and Screen Recording iPads back to the main room and plug in / set for next time.